

Web Application Project Proposal (MERN Stack)

1. Project Title

DonateBridge – Community Item Donation & Pickup Platform

2. Problem Statement

Many households have usable items, clothes, books, furniture, and small appliances, that they no longer need but that could genuinely help someone else. At the same time, NGOs and low-income families are often actively looking for exactly these kinds of items. Right now, there is no simple, direct way for these two sides to find each other.

- **Who is affected:** Individual donors, who have items to give away but no easy channel to reach an NGO or a family in need, other than word of mouth or dropping items at a single collection point; NGOs, who spend time manually coordinating with donors over phone calls or social media posts to figure out what is available and where; and needy families, who often depend entirely on the NGO to know what has been donated, with no visibility of their own into what items exist.
- **Current challenges or limitations:** Donations are usually announced informally through Facebook posts or word of mouth, so many potential donors and recipients never find out about them. NGOs have no organized record of which items were offered, which were already picked up, and which are still pending, so items are sometimes forgotten or collected twice. There is also no clear approval step, so an NGO cannot easily confirm a donation before sending someone to collect it, and pickup scheduling is handled through ad-hoc phone calls with no written record.
- **Why existing solutions are insufficient:** General marketplace apps (Facebook Marketplace, classifieds sites) are built for buying and selling, not donations, so they have no concept of a verified NGO, a pickup request, or a collection schedule. Large charity platforms mostly focus on monetary donations rather than physical items, and none of them are built for the simple, local, item-based donation flow that small NGOs and individual donors actually need.
- **Impact of the problem:** Usable items end up thrown away or left unused instead of reaching someone who needs them, NGOs waste time on manual coordination instead of fieldwork, and donors who want to help often give up simply because there is no convenient way to do so.

3. Proposed Solution

DonateBridge is a web application that connects individual donors with NGOs through a simple item-listing and pickup-request workflow. Donors post items they want to give away, NGOs browse available items and request pickup, and an admin panel lets NGOs (and the platform admin) verify requests and schedule collection, all tracked through a clear status lifecycle for every donation (Listed → Requested → Scheduled → Collected).

- **Core idea:** A centralized platform where donated items, pickup requests, and collection schedules are all tracked in one place, so both donors and NGOs always know the current status of an item instead of relying on phone calls or memory.
- **Target users:** Donors, who can list items with a description, category, and photo, and track what happens to them; NGOs, who get a dashboard to browse listed items, send pickup requests, and manage their own collection schedule; and an admin (representing the NGO's staff or the platform owner), who verifies pickup requests and confirms the final schedule.
- **Values the system provides:** Gives donors a simple way to list an item in a few steps instead of posting on social media, gives NGOs a searchable, centralized view of everything currently available, prevents confusion over which items are already claimed through a clear status system, and keeps a permanent record of past donations that NGOs can refer back to.

- **How it fits daily use:** A donor lists an item (e.g., a bag of clothes or a study table) with a short description, category, and condition. NGOs browse or filter the listing page and send a pickup request for items they want. The item's status changes to Requested, and the admin panel lets the NGO confirm the request and set a pickup date and time, at which point the status becomes Scheduled. Once the item is physically collected, the admin marks it as Collected, closing the loop and keeping a permanent record for both the donor and the NGO.

4. Features & Functional Requirements

The main features of the system are:

- **Public item browsing (no login required):** any visitor can view the list of currently available donated items, including photo, category (clothes/books/furniture/other), condition, and a short description, so people can see what is being offered before signing up.
- **Donor registration and login:** donors create an account (name, email, phone, password) and log in before listing an item; authentication is handled using JWT.
- **Item listing by donors:** a logged-in donor can add a new item with a title, category, condition, description, and photo, and can edit or remove their own listing as long as it has not yet been requested.
- **Donor listing history:** donors can view all the items they have posted along with the current status of each one (Listed, Requested, Scheduled, or Collected).
- **Role-based authentication and authorization:** two main roles, Donor and NGO/Admin, each with their own login and permissions; NGO/Admin accounts are created by the platform admin, while donor accounts are self-registered.
- **NGO browsing and pickup requests:** logged-in NGO users can browse all available items, filter by category, and send a pickup request for an item, which changes its status from Listed to Requested and links it to that NGO.
- **Admin verification panel:** the admin can view all incoming pickup requests, verify that a request is genuine, and either approve or reject it before scheduling; rejected requests return the item to Listed status so other NGOs can request it.
- **Pickup scheduling:** once a request is approved, the admin sets a pickup date and a collection address/time slot, moving the item's status to Scheduled; this schedule is visible to both the donor and the requesting NGO.
- **Automated item status workflow:** every item automatically moves through Listed → Requested → Scheduled → Collected as the donor lists it, an NGO requests it, the admin schedules it, and the pickup is finally confirmed, without needing a separate manual tracking sheet.
- **Collection confirmation:** after the item is physically picked up, the admin marks it as Collected, which removes it from the active browsing list and moves it into the permanent donation history.
- **Search and filter for items:** both donors (checking similar past donations) and NGOs (when browsing) can filter items by category, condition, or status.
- **Access control between roles:** donor accounts cannot access the admin verification panel or see other donors' contact details, and NGO accounts can only see items and requests relevant to their own activity, ensuring each role only sees what is appropriate for them.
- **Donation history and records:** every collected donation is stored permanently with donor, NGO, item, and pickup details, giving the admin a simple record of the NGO's overall donation activity over time.